

Mobile Networks Project - Sub-THz Communications

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1 Part 1: PHY Specification in THz(Clause 14)

1.1 Understanding

Terahertz (THz) communications extend wireless systems beyond millimeter-wave frequencies to support ultra-high data rates. IEEE Std 802.15.3d-2023 [1]. defines a PHY layer operating in the THz band, enabling data rates up to and beyond 100 Gbit/s. This part of the report, presents a structured overview of the THz PHY specification defined in Clause 14 of the standard.

1.2 General requirements

1.2.1 Modes of Operation

A compliant THz PHY shall implement at least one of the following PHY modes:

- Single Carrier (THz-SC PHY) Mode: For high-performance, long-range links (up to hundreds of meters) requiring complex modulation and high data rates.
- On-Off Keying (THz-OOK PHY) Mode: For ultra-low-power, short-range devices (centimeters) that cannot handle the hardware complexity of SC mode.

1.2.2 RF Power Measurements and Channelization

All RF power measurements are corrected for antenna gain based on manufacturer specifications. Supported bandwidths range from 2.16 GHz to 69.12 GHz in multiples of 2.16 GHz. The 2.16 GHz frequency band is adopted to ensure compatibility with existing millimeter-wave and Wi-Fi 7 hardware implementations. Default channels operate at 4.32 GHz bandwidth.

Spectrum range	Default CHNL_ID
252–325 GHz	41
356–450 GHz (lower part)	137
356–450 GHz (lower part)	139

Table 1: Spectrum ranges and default channel IDs

The Lower part of the frequencies operating in the 4.32 GHz Bandwidths (CHNL_ID 137 and 139) are dedicated for experimental purposes and are subjected to molecular absorption.

1.2.3 Transmission PSD Mask:

The "containment zone" - It ensures that while a device is talking on its own channel, it isn't "shouting" into neighboring frequencies and causing interference. PSD masks define channel boundaries and limit adjacent-channel interference.

Three regions are defined:

The In-band region, where the signal power is concentrated within the allocated bandwidth.

The Transition region, which provides a controlled spectral roll-off to limit adjacent-channel interference.

The Out-of-band/Stop-band region, where strict attenuation requirements are enforced to prevent interference with other systems. (Figures 1 and 2)

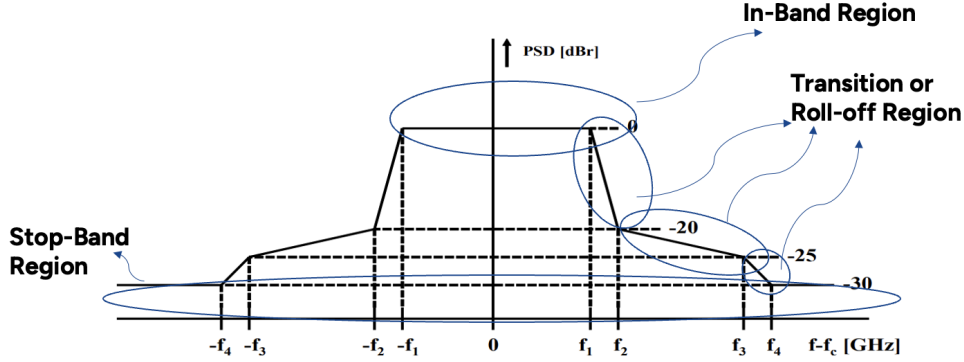


Figure 1: Transmisson PSD mask for THZ-SC-PHY Mode

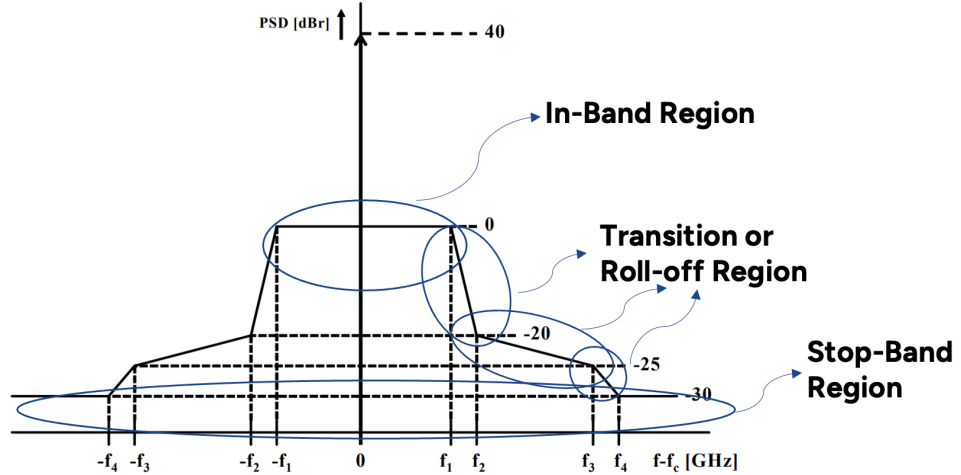


Figure 2: Transmisson PSD mask for THZ-OOK-PHY Mode

1.2.4 THZ PHY Management

The PHY PIB(PAN Information Base) defines the managed objects required to control the THz PHY layer of a PRDEV(Pairnet Device). It specifies the supported PHY types and modes, regulatory domains, data rates, channel capabilities, and frame parameters. The maximum frame size is 2,099,200 octets, the maximum transfer unit is 2,097,152 octets, and the minimum fragment size is 2,048 octets. All PHY parameters are configurable via read/write access.

1.3 THZ SC PHY Mode

The physical layer (PHY) specification is fundamentally defined by three essential components, namely channelization, modulation, and coding, which collectively determine the system's spectral efficiency, robustness, and overall performance.

1.3.1 Channelization and Modulation

Channelization follows the default channel definitions defined in section 1.2.2.

The THz-SC PHY is designed for high-performance and longer-range links, supporting complex modulation schemes. Modulation options include $\pi/2$ -BPSK, $\pi/2$ -QPSK, $\pi/2$ -8PSK, and $\pi/2$ -16QAM (figure 3). These constellations enable a trade-off between spectral efficiency and robustness.

Modulation	Bits per Symbol	Primary Use Case
$\pi/2$ -BPSK	1	Robust transmission with very low PAPR, suitable for control signaling and low data rate links
$\pi/2$ -QPSK	2	Balanced trade-off between data rate and spectral efficiency; standard THz operating mode
$\pi/2$ -8PSK	3	High data rate transmission where bandwidth efficiency is a priority

Table 2: THz-SC PHY Modulation Schemes

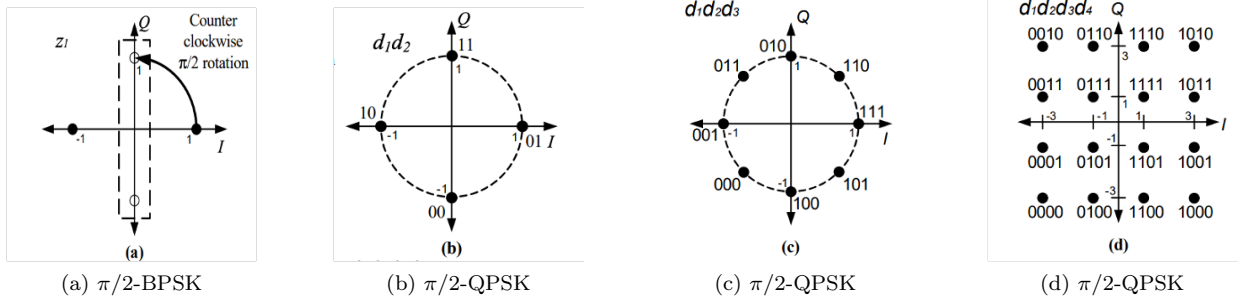


Figure 3: Constellation diagrams for THz-SC modulation schemes.

1.3.2 Modulation and Coding (MCS) Parameters

Forward error correction (FEC) is achieved using two mandatory LDPC codes: a rate-14/15 LDPC(1440,1344) code and a rate-11/15 LDPC(1440,1056) code. Key parameters include a block length of 64 chips, a mandatory pilot word (PW) length of 8 chips, and an optional zero-length PW. Although data rates without PW are higher, the PW significantly improves signal reliability and synchronization accuracy (figure 4).

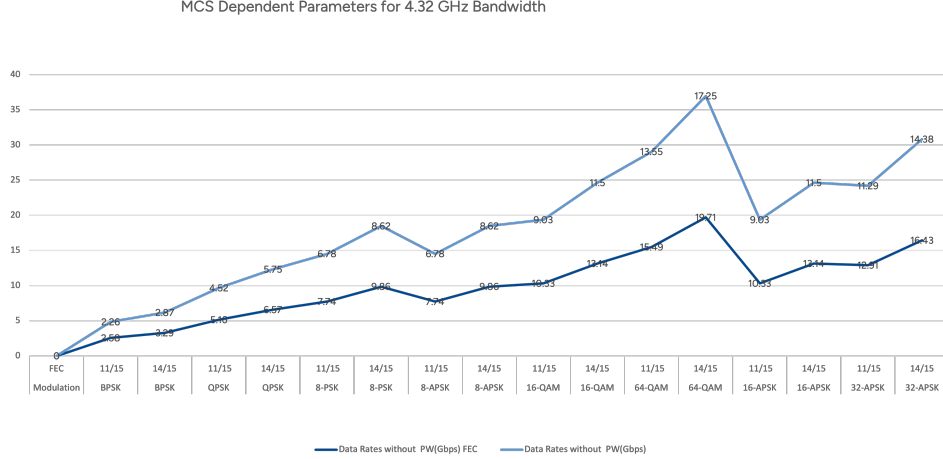


Figure 4: Date rates for Different Modulation Techniques at Different FEC rates

1.3.3 THz-SC Mode MAC Layer Overview

The THz-SC PHY frame consists of a preamble, header, and payload. The preamble supports receiver functions such as automatic gain control, timing acquisition, frequency offset estimation, and channel estimation. The payload construction process includes scrambling, encoding, insertion of stuff bits, constellation mapping, block formation, and periodic insertion of pilot patterns.

Transmitter specifications define error vector magnitude (EVM) limits for each modulation and FEC combination, while the transmitter frequency tolerance is specified by the MAC parameter *pPHYClockAccuracy* of $\pm 30 \times 10^{-6}$. Receiver specifications include error rate criteria after error correction and receiver sensitivity thresholds. Timing parameters cover receive-to-transmit turnaround time, transmit-to-receive turnaround time, inter-frame spacing, and channel switching time.

1.4 THz-OOK Mode Overview

The THz-OOK PHY targets ultra-low-power, short-range applications with reduced hardware complexity. It uses simple OOK modulation and mandatory Reed–Solomon (224/240) coding, with optional LDPC codes of rates 14/15 and 11/15. The OOK PHY follows the THz SC Mode channelization, timing-related parameters, frame structure, payload construction, and stuff bit handling, and is optimized for energy-efficient operation.

1.5 Overall Discussion and Recent Studies

IEEE 802.15.3d represents a significant milestone in standardizing THz communications, enabling data rates of 100 Gbit/s and beyond. Recent studies focus on compatible channelization schemes and advanced hardware implementations, such as broadband modified uni-traveling carrier (MUTC) photodiodes, to realize practical THz transceivers.

The dual-PHY approach (SC and OOK) allows the standard to address a wide range of use cases, from high-throughput point-to-point links to short-range, ultra-low-power devices. This flexibility makes IEEE 802.15.3d highly relevant for future 6G systems and emerging applications in wireless personal and local area networking.

1.6 Conclusion

This part of the report presented a detailed discussion of the THz PHY specification defined in IEEE Std 802.15.3d-2023, with emphasis on RF channelization, PSD masks, PHY management, and PHY mode specifications. By addressing both performance-intensive and energy-constrained scenarios, the standard lays a strong foundation for next-generation THz wireless communications.

2 Part 2: Deterministic Setup

2.1 Task 1: Gain Calculation

The first part of the project is to calculate the gain of a planar antenna array of size X cm per side with respect to frequency and size. The element spacing is $\lambda/2$ and the elements are modeled as point sources. The envisioned geometry is depicted in Figure 5.

To calculate the gain of the array, the following formula was used:

$$G = \eta D = \eta 4\pi \frac{A_{\text{eff}}}{\lambda^2} \quad (1)$$

Where, $A_{\text{eff}} = \alpha A_{\text{phy}}$ is the effective aperture of the array, A_{phy} is the physical aperture of the array and $\alpha \in [0, 1]$ is the aperture efficiency. In this assignment, ideal aperture efficiency was assumed, so $\alpha = 1$. Also, the elements are assumed to be ideal radiators, meaning no losses, enabling us to set $\eta = 1$ as well. For the given two-dimensional array, the total size is $A_{\text{phy}} = (\frac{N\lambda}{2})^2$.

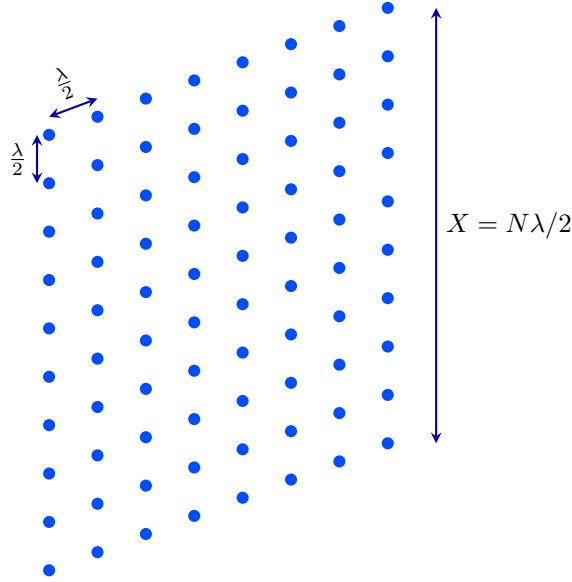


Figure 5: Array representation

2.2 Gain vs frequency

To calculate the maximum gain possible at frequency f given side length X , the following procedure was executed: At each frequency point f_i , we first calculate the number of elements per dimension as: $N = \lfloor \frac{2X}{\lambda_i} \rfloor$. Then the aperture is calculated as $A_{\text{eff}} = (\frac{N\lambda_i}{2})^2$. This procedure suggests that each frequency point is considered to be the design frequency of the array - allowing for $\lambda/2$ spacing. The results suggest the maximum theoretical gain that can be achieved by an array, rather than the performance of a specific antenna.

Figure 6 showcases the dependency of gain in dBi with respect to the frequency in GHz for various values of the array side length X in cm. Larger values of X lead to higher gain across the whole frequency range because the term $\lfloor \frac{2X}{\lambda_i} \rfloor$ allows for more element to be placed in the array, leading to larger effective aperture. Another point that should be highlighted is that smaller physical sizes showcase less discrete points in the graphs, which is again a result of the reduced assigned dimensions. For example, take the case $X = 1\text{mm}$. Then, at $f = 150\text{GHz}$, $X = \lambda/2$, allowing for an array of 2-by-2 elements, while at $f = 450\text{GHz}$, $X = 1.5\lambda$ allowing for an array of 4-by-4 elements. In contrast, $X = 1\text{cm}$ allows for 20-by-20 and 40-by-40 elements in the same frequencies, with all the intermediate values in between.

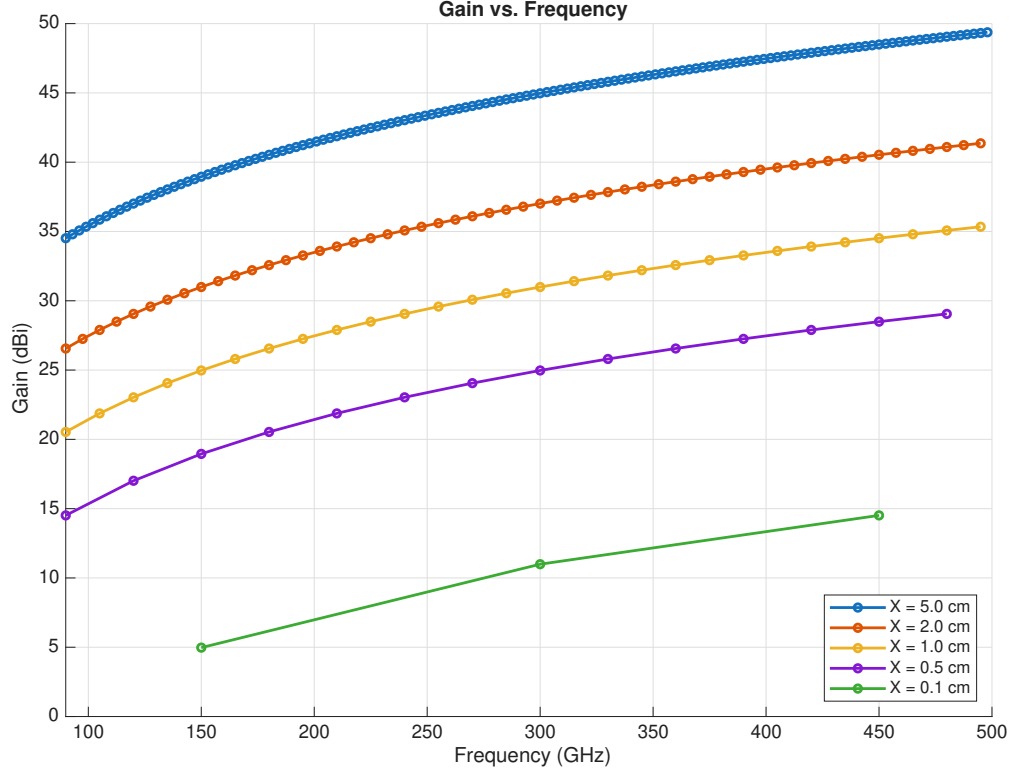


Figure 6: Gain vs Frequency

2.3 Gain vs Array size

The second plot refers to the obtained gain in dBi plotted against the physical size X of the array for different design frequencies ranging from 100 GHz up to 1THz. Here we observe that higher design frequency results in higher gain for the same assigned physical space. This is because the gain is inversely proportional to the square of the wavelength.

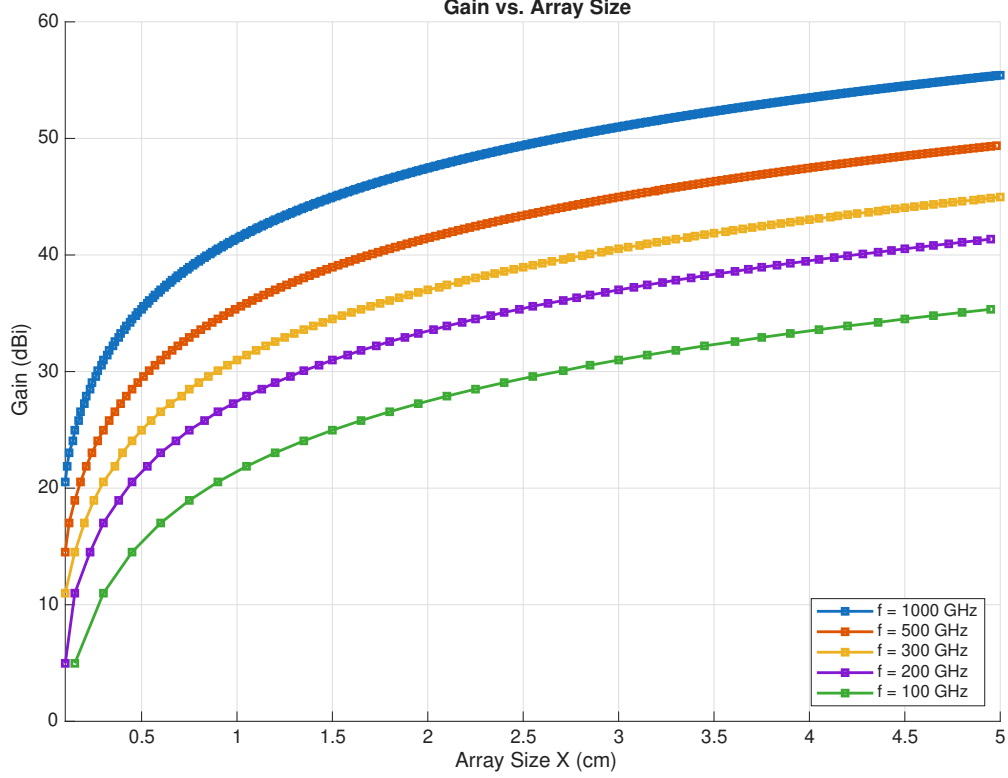


Figure 7: Gain vs Array Size

2.4 Task 2: Path Loss

In this task, the goal is to calculate the total path loss between two points spaced at distance d , assuming the operating frequency is somewhere in the sub-THz and THz bands. The methodology is directly adopted from the reference paper in the instructions of the project [2].

The total path loss is given as a combination of total free space propagation with the effects of molecular absorption caused by atmospheric gasses, mainly oxygen and water vapor. The expressions for these are:

$$A_{\text{FSPL}} = \left(\frac{4\pi f d}{c} \right)^2 \quad (2)$$

$$A_{\text{abs}} = e^{k(f)d} \quad (3)$$

$$A_{\text{total}} = \left(\frac{4\pi f d}{c} \right)^2 e^{k(f)d} \quad (4)$$

The term $k(f)$ is the molecular absorption coefficient which captures the effect of the atmosphere. The value of the absorption coefficient depends on the mixture of elements in the medium of propagation and their ratio. For this reason, the effect of water vapor concentration will be studied later. All data for $k(f)$ was obtained through the HITRAN database (<https://hitran.org/mtckd/>) and loaded using the HITRAN Application Programming Interface (HAPI). In order to plot the results, the obtained attenuation is converted into dB:

$$A_{\text{dB}} = 20 \log_{10} \left(\frac{4\pi f d}{c} \right) + 10k(f)d \log_{10}(e) \quad (5)$$

As a means to ensure correct handling of the data and compare different methods for modeling the attenuation caused by the atmospheric gasses, the methodology above was tested against the ITU-R 676 recommendation [3]. In that case, the attenuation per kilometer is calculated in a line-by-line fashion, by summing all the individual resonance lines through the expression:

$$\gamma = 0.1820 N''(f) \quad (6)$$

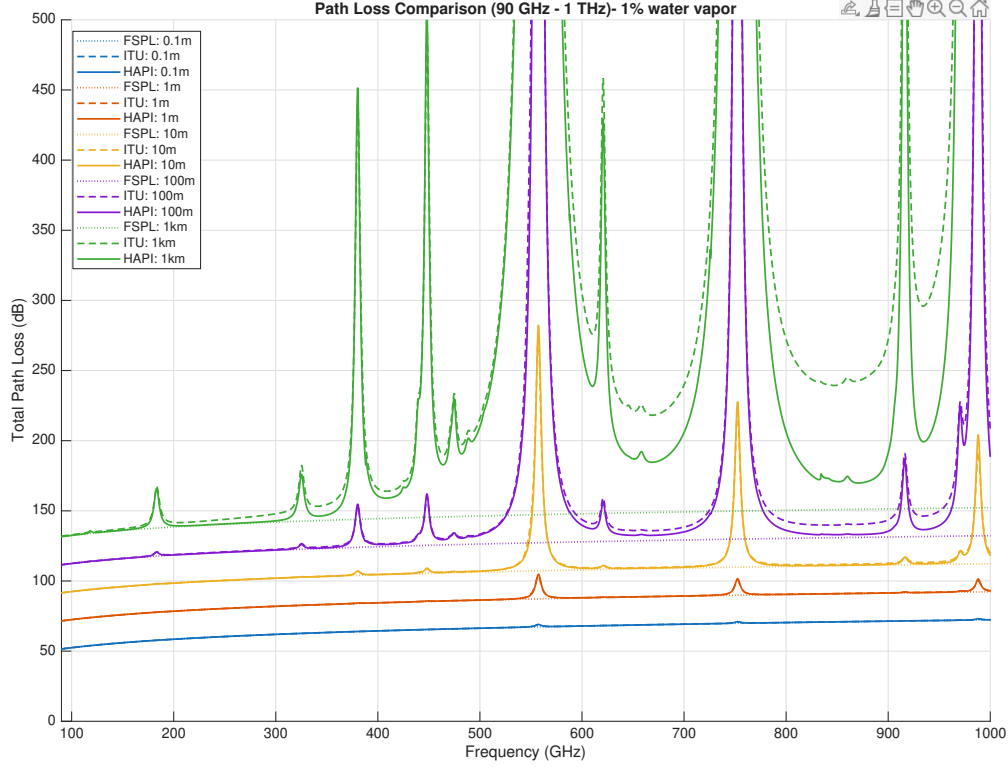


Figure 8: Total attenuation, Method comparison

Where $N''(f)$ is the imaginary part of the frequency dependent complex refractivity. The details for obtaining the values of $N''(f)$ are found in the standard, but what should be mentioned is the fact that the specific attenuation is affected by the choice of water vapor density ρ and partial pressure $e = \frac{\rho T}{216.7}$, where T is the temperature in Kelvin. To ensure consistency between the two methods, the water vapor density and partial pressure have to be converted into a ratio to use as input in HAPI to obtain the correct mixture in the atmosphere. We can obtain the mixing ratio of water vapor molecules in the atmosphere by calculating the ratio of pressures e/P , where P is the total barometric pressure.

These results are shown in figure 8. We observe very good consistency for smaller distance values, up to hundreds of meters. For bigger distances as well as higher frequencies, the radiative transfer theory approach proposed by the authors of [2] provides more accurate results, so for the rest of this report we shall use it to model the propagation mechanisms.

In figure 9 the total attenuation in dB is plotted against frequency in GHz, covering the whole sub-THz band (90-1000 GHz), as well as distance in meters, ranging from 0 up to 1km. We observe large spikes in certain frequencies, mainly at 550, 750 and 1000 GHz. These are caused by water molecules' resonant frequencies in the sub THz band, resulting in increased values of the molecular absorption coefficient. One more observation we can make is that these spikes, although narrow at small distances, tend to spread once we increase the distance, affecting more bandwidth. This is because the attenuation by molecular absorption scales exponentially with distance, and thus impacting long range links more severely. The effect of water molecule concentration in the atmosphere was studied next. In figure 10, three different scenarios have been simulated, taken from ITU-R P.835-7 recommendation [4]. They correspond to standard conditions (15°C , $1\% \text{H}_2\text{O}$ - Green Curves), humid summer (28°C , $2.5\% \text{H}_2\text{O}$ - Red Curves) and dry winter conditions (-1°C , $0.4\% \text{H}_2\text{O}$ - Blue Curves). At small distances, the behavior is similar, but once we increase the range into 100 m or even 1k m we can see clear separation in the levels of attenuation. The environment starts to affect the bands that lie outside of the resonant peaks of water vapor. Observe that for the red dotted curve (Summer-1 km range) above 450 GHz the attenuation is constantly above 200 dB. These results indicate a clear dependency of channel conditions in the humidity of the environment, which should be handled when

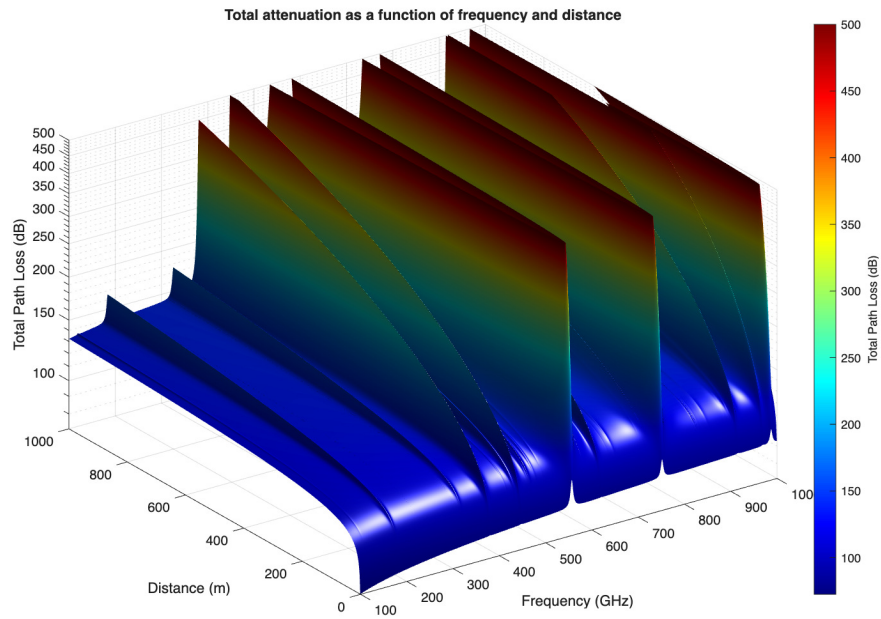


Figure 9: Total attenuation against frequency and distance

designing sub-THz links, especially in longer ranges.

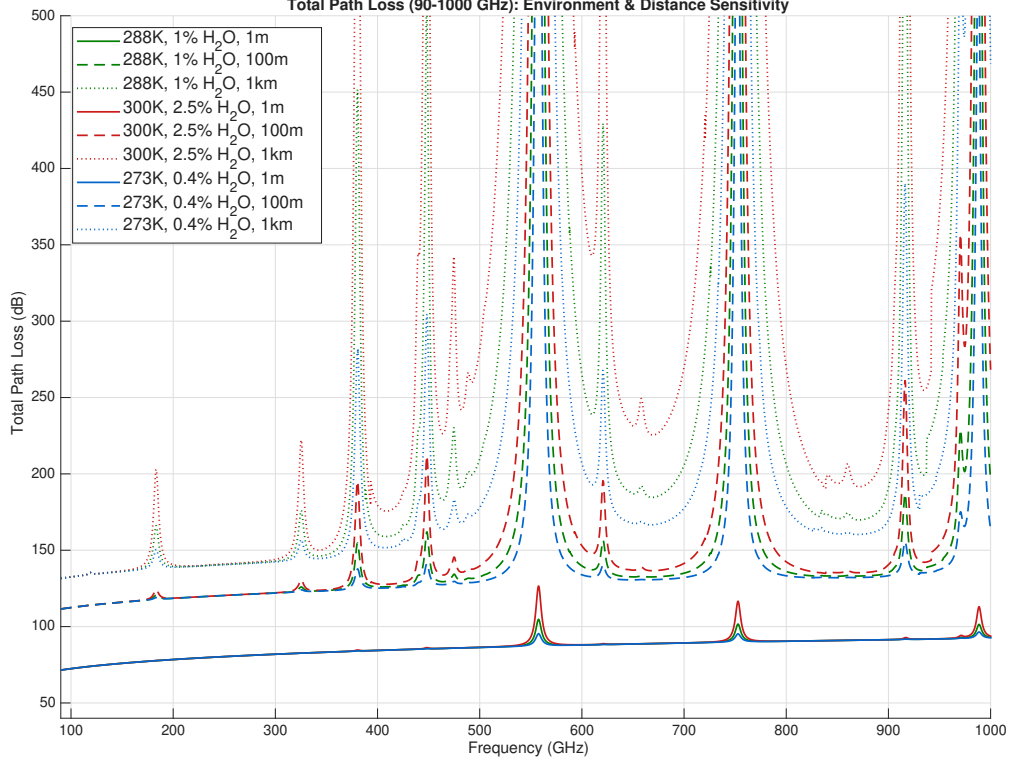


Figure 10: Attenuation for different atmospheric conditions

2.5 Task 3: Capacity calculation

At this point, we aim to combine the results of the two previous tasks to compute the capacity of the sub-THz band. As we just discussed the channel cannot be modeled as frequency-flat so we introduce the capacity equation for frequency selective channels. The total bandwidth is split into slices where the attenuation, gains, power transmitted are assumed flat. The expression is the following:

$$C(d) = \sum_i \Delta f_i \log_2 \left(1 + \frac{P_t(f_i) G_T(f_i) G_R(f_i) A^{-1}(f_i, d)}{k T_0 \Delta f} \right) \quad (7)$$

Where the bandwidth of each channel is $\Delta f = 5\text{MHz}$ and the denominator corresponds to noise power with equivalent temperature $T_0 = 296\text{K}$. In terms of transmitted power, there are two strategies employed. First, a simple uniform power allocation was used, where each subchannel got the same power $P_t(f_i) = P_t / N_{\text{total}}$. Next, the water-filling algorithm was employed where subchannels with lower perceived noise power are prioritized in terms of power allocated. In the previous expression, if we set $P_n(f_i) = \frac{k T_0 \Delta f A(f_i, d)}{G_T(f_i) G_R(f_i)}$ then, the water-filling algorithm takes as inputs the total transmitted power P_t and the vector containing the values of $P_n(f_i)$ and returns the vector $P_t(f_i)$ that maximizes the sum rate C .

In figure 11, the results in capacity (Tbps) are plotted against distance in meters. Three different cases have been studied. First, the whole sub-THz band is utilized (90-1000 GHz, Blue curves), then, the bandwidth is reduced to only the bottom half of the previous band (90-500 GHz, Purple curves) and lastly, the THz spectrum is utilized (1-10 THz, orange curves) to experiment with higher frequencies. We can observe that for small distances, under 0.1 m, utilizing more bandwidth leads to better capacity, since the attenuation by molecular absorption is negligible. As the distance increases, we observe that the higher frequency bands suffer, resulting in lower capacity. This is because, above 500 GHz, molecular absorption peaks are very frequent, making the extra bandwidth non-beneficial. As an example, take a look at the blue and purple solid lines, corresponding to the water-filling algorithm for sub-THz and bottom half of sub-THz band respectively. Above several meters of distance they behave identically, meaning that the upper band (500-1000 GHz) is unutilized.

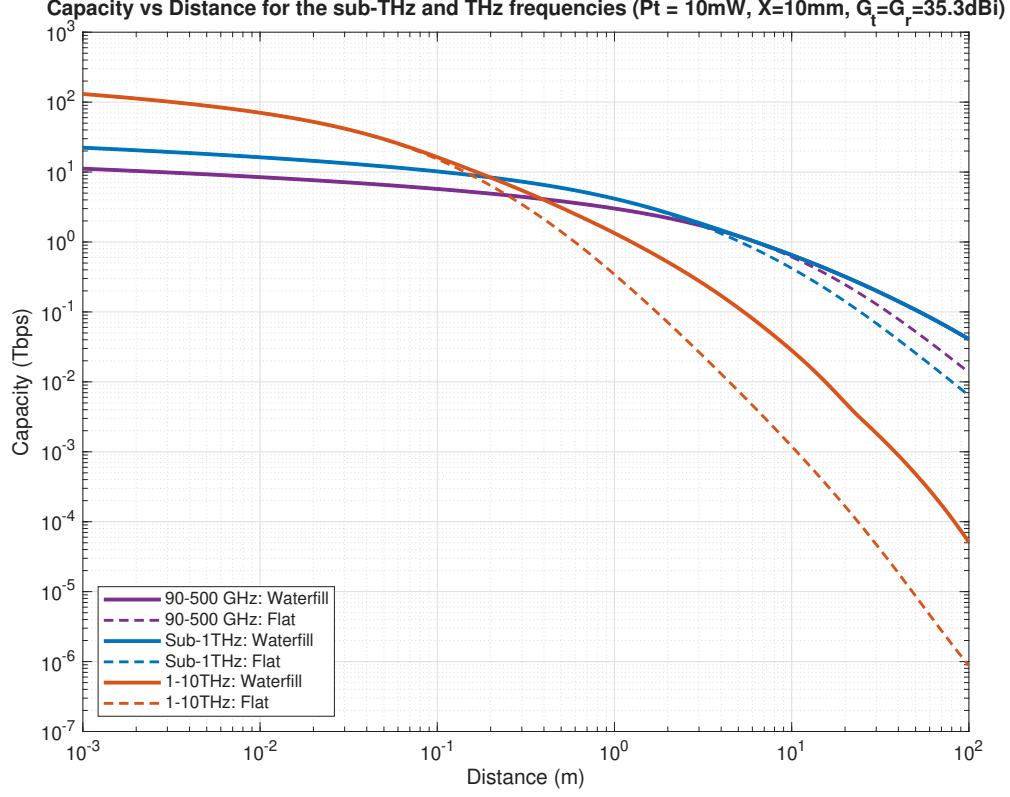


Figure 11: Capacity vs distance - sub-THz band

2.6 Task 4: Backhaul Link

In this task, we combine the three previous modules to model a backhaul link. The goal is to determine the size of the transmit and receive arrays from task 1, in order to achieve a specific capacity requirement. For this scenario, the bandwidth utilized is 10 GHz. This enables us to assume the channel attenuation as well as the gains of the antennas are constant throughout the whole channel. This way, we can solve the capacity equation with respect to the gain, assuming identical antennas in the transmitter and receiver.

$$C = B \log_2 \left(1 + \frac{P_T G^2 A^{-1}}{k T_0 B} \right) \Rightarrow G = \sqrt{(2^{C/B} - 1) \frac{k T_0 B A}{P_T}} \quad (8)$$

Then, solving for the array side length we obtain: $X = \lambda_c \sqrt{\frac{G}{4\pi}}$. Then, the minimum obtainable size of the array can be calculated by determining the number of elements that fit into the assigned size.

$$N_{\min} = \lceil \frac{2X}{\lambda_c} \rceil, X_{\min} = \frac{N_{\min} \lambda_c}{2} \quad (9)$$

Where λ_c denotes the wavelength at the design frequency.

The results are shown in figure 12. The required side array length in mm is plotted against the target capacity in Gbps for three different design frequencies at three different distances. On all curves, we observe an exponential increase after a certain point (300-400 Gbps depending on the case) which is caused by the logarithmic increase of capacity with respect to Signal-to-Ratio. No matter how much the SNR term inside the logarithm is increased, after a point, the increase in capacity will be negligible. In addition, increasing the physical size beyond logical values defeats the purpose of small form factors, and the number of elements becomes too large to handle. We also observe that at bigger distances, the difference in required size of the array between the three cases shrinks. This is happening because higher frequencies experience greater attenuation at large distances as we discussed in the previous tasks.

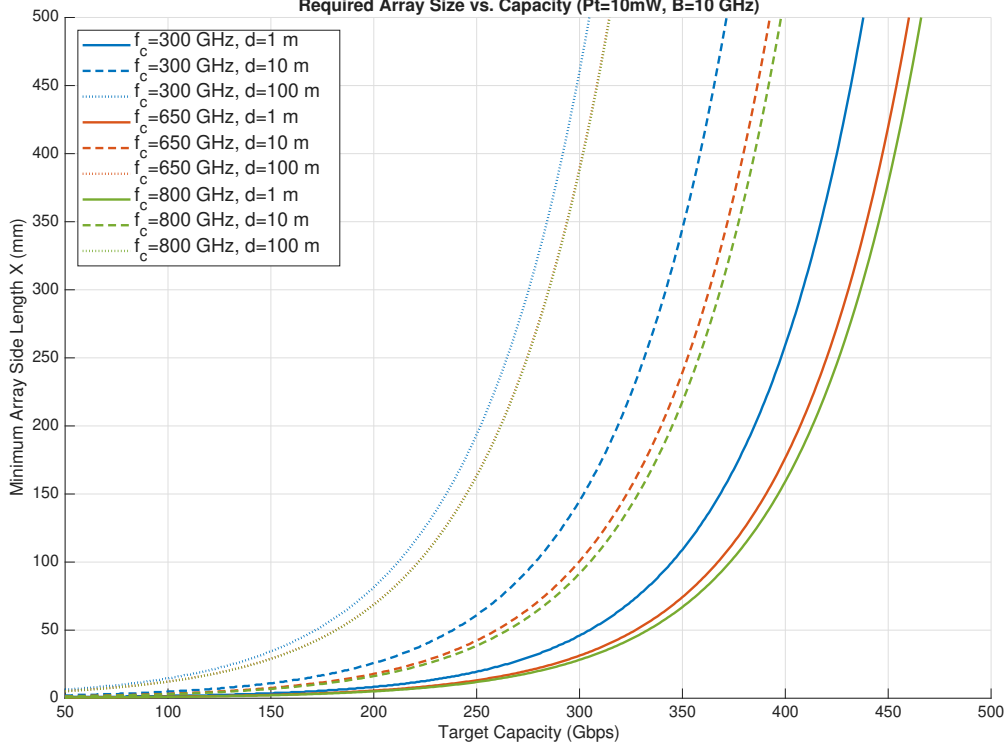


Figure 12: Required array side length vs target capacity

3 Part 3: Stochastic Beam Alignment

3.1 System and Antenna Model

This section describes the system model used to evaluate the initial beam alignment delay in sub-THz links. In the 2D scenario, the transmitter (Tx) is located at the origin $(0, 0)$ and the receiver (Rx) is placed at $(0, d)$. The Rx is assumed to remain perfectly aligned towards the Tx throughout the experiment. Therefore, the alignment uncertainty exists only at the Tx side. The Tx initial steering direction is modeled as a uniformly distributed random variable,

$$\theta_0 \sim \mathcal{U}(0, 360). \quad (10)$$

Both Tx and Rx are modeled using an ideal directional antenna with beamwidth α (in degrees), following a sector shaped pattern where the gain is constant inside the main lobe and zero elsewhere. Side lobes and back lobes are neglected. In the 2D case, for a Tx steering direction θ_{Tx} , the Tx beam covers the angular interval

$$\theta \in \left[\theta_{Tx} - \frac{\alpha}{2}, \theta_{Tx} + \frac{\alpha}{2} \right] \quad (11)$$

and beam alignment is completed when the Rx direction θ_{Rx} lies inside this interval.

Time is represented in beam probing attempts. Each beam steering attempt incurs a fixed cost of 1 ms. Therefore, the beam alignment time is defined as

$$T_{\text{align}} = N_{\text{attempts}} \times 1 \text{ ms} \quad (12)$$

where N_{attempts} is the number of attempted beam directions until successful overlap.

In the 3D case, the Tx beam is modeled as an ideal cone with beamwidth α . The beam steering directions are represented by unit vectors, where $\mathbf{u}_{Tx}$ denotes the Tx steering direction and $\mathbf{u}_{Rx}$ denotes the direction towards the receiver. For Tx steering direction $\mathbf{u}_{Tx}$ and Rx direction $\mathbf{u}_{Rx}$, alignment occurs when the Rx direction lies within the Tx cone. The antenna is modeled as a cone with opening angle α , and alignment

occurs when the angular separation satisfies

$$\angle(\mathbf{u}_{Tx}, \mathbf{u}_{Rx}) \leq \frac{\alpha}{2}. \quad (13)$$

3.2 Beam Alignment Strategies

This section describes the beam-search strategies evaluated in both the 2D and 3D alignment scenarios.

3.2.1 S1: Clockwise Sequential Scanning

In Strategy 1, the transmitter starts from its initial steering direction θ_0 and performs a sequential scan by rotating the beam clockwise with a fixed angular resolution of 1° per step.

3.2.2 S2: Counter-Clockwise Sequential Scanning

Strategy 2 is identical to Strategy 1 except that the beam is rotated counter-clockwise with a step size of 1° .

3.2.3 S3: Random Probing with Repetition

In Strategy 3, the transmitter probes directions randomly. At each attempt, the transmitter chooses a random beam direction from all possible directions. Since each probe is independent, the same direction may be selected multiple times, which can lead to redundant attempts.

3.2.4 S4: Random Probing without Repetition

Strategy 4 reduces redundancy compared to Strategy 3 by avoiding repeated beam directions. This is achieved by storing the previously tested beam directions (in 2D) and using angular bins on the sphere (in 3D), and ensuring that each direction is selected at most once.

3.2.5 S5: Variable-Step Structured Scanning

In Strategy 5, the transmitter performs structured scanning with a step size proportional to the beamwidth, which reduces the effective number of probed directions. The step size is defined as

$$\Delta = \beta\alpha \quad (14)$$

where β is a scaling factor. In 2D, the transmitter scans sequentially starting from a random initial direction θ_0 with step size Δ . In 3D, the scan is performed over a pre-defined beam grid covering the sphere, starting from a random index.

3.2.6 S6: Hybrid Grid-Based Random Scanning

Strategy 6 combines structured coverage with randomised probing. In the 2D case, the same set as Strategy 5 is used, but the order of probing is randomised to reduce dependency on scan ordering. In the 3D case, a beam grid covering the sphere is generated, and all grid points are randomly shuffled to create a randomised scan order without repetition.

3.3 Simulation Methodology

Monte Carlo simulations were conducted to estimate the initial beam alignment delay for each beam search strategy. For a given beamwidth α , the transmitter initial steering direction was randomised, and the corresponding beam alignment time was recorded as the number of beam probing attempts required to achieve alignment. Since each probing attempt is assumed to take 1 ms, the alignment time in milliseconds is numerically equal to the number of attempts.

In addition to the mean, the probability density function (PDF) of the alignment time was evaluated. This allows comparison of strategy reliability and worst-case behavior.

The simulations were performed for both 2D and 3D settings. In the 2D case, beam directions were represented by a single azimuth angle $\theta \in [0, 360)$, while in the 3D case beam directions were represented as unit vectors on the sphere. Random beam directions in the 3D case were generated uniformly on the unit sphere using the standard sphere point picking method [5]. In the 3D structured strategies (S5 and S6), a beam grid covering the sphere was generated, and probing was performed over this finite set of beam directions.

3.4 Results

3.4.1 2D Results: Mean Alignment Time

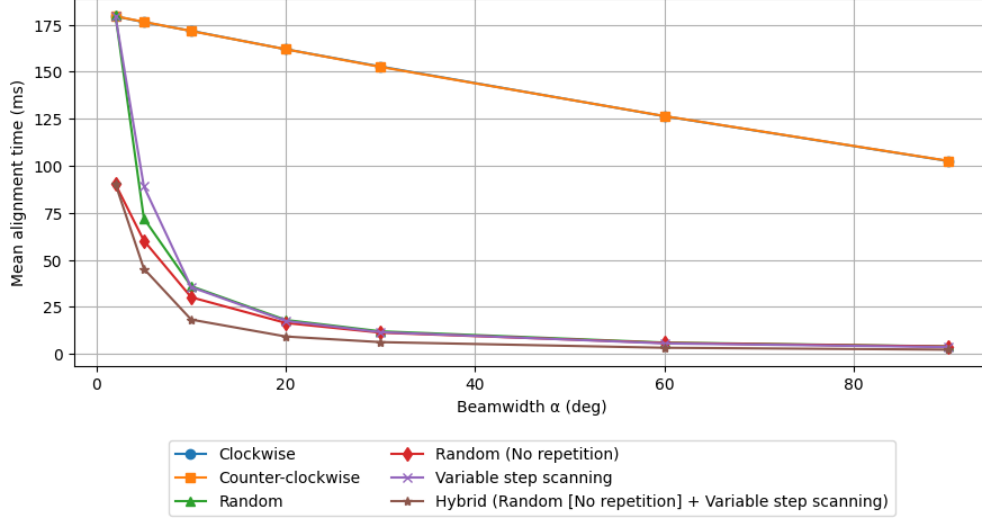


Figure 13: Mean Alignment Time vs Beamwidth (2D)

We evaluated all six beam search strategies in the 2D scenario for beamwidth values $\alpha \in \{2^\circ, 5^\circ, 10^\circ, 20^\circ, 30^\circ, 60^\circ, 90^\circ\}$, using $N_{\text{trials}} = 10^6$ Monte Carlo trials per beamwidth. Figure. 13 shows the mean alignment time as a function of α . The sequential scanning strategies (S1 clockwise and S2 counter-clockwise) shows nearly identical mean performance, confirming that scanning direction does not affect average alignment delay. Compared to sequential scanning, random probing strategies significantly reduce alignment time. Strategy 4 (random probing without repetition) consistently outperforms Strategy 3 due to reduced redundant probing. The hybrid strategy (S6) achieves the lowest mean alignment time across all beamwidth values, particularly for narrow beams.

3.4.2 2D Results: Probability Density Function

We next evaluate the distribution of alignment delay using the probability density function (PDF). PDF results for the 2D scenario are presented for $\alpha = 10^\circ$ using $N_{\text{trials}} = 10^6$ Monte Carlo trials. As shown in Figure. 14, the sequential scanning strategies (S1 and S2) produce an almost uniform distribution over a wide range of alignment times, indicating that alignment may occur at any point during the scan depending on the initial orientation. This results in high variability and long-tail behavior, which corresponds to slow worst-case alignment.

In contrast, as shown in Figure. 15, the randomised strategies (S3 and S4) concentrate most of the probability mass at smaller alignment times and exhibit a decaying tail, implying faster typical performance. Strategy 5 (variable-step scanning) shows a bounded distribution with approximately uniform probability across a limited time range, since it sequentially scans a set of beam directions with step size proportional to α . Finally, the hybrid strategy (S6) provides the highest probability of fast alignment, with PDF strongly concentrated at low alignment times. Overall, the PDF analysis confirms that sequential scanning is highly

unreliable in worst-case delays, whereas structured and hybrid strategies improve predictability and significantly reduce extreme alignment delays in 2D.

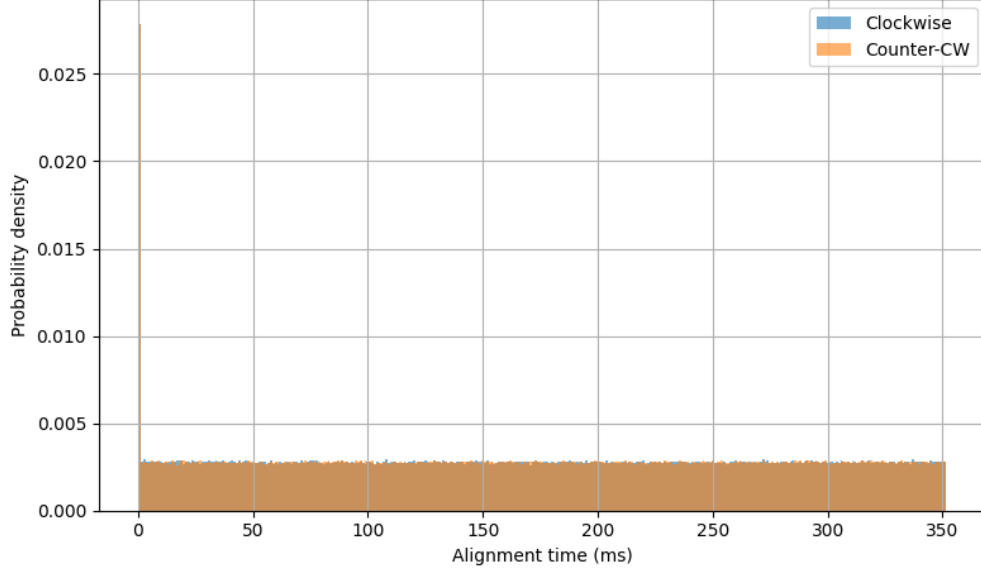


Figure 14: PDF of Beam Alignment Time in 2D for S1 & S2 ($\alpha = 10^\circ$)

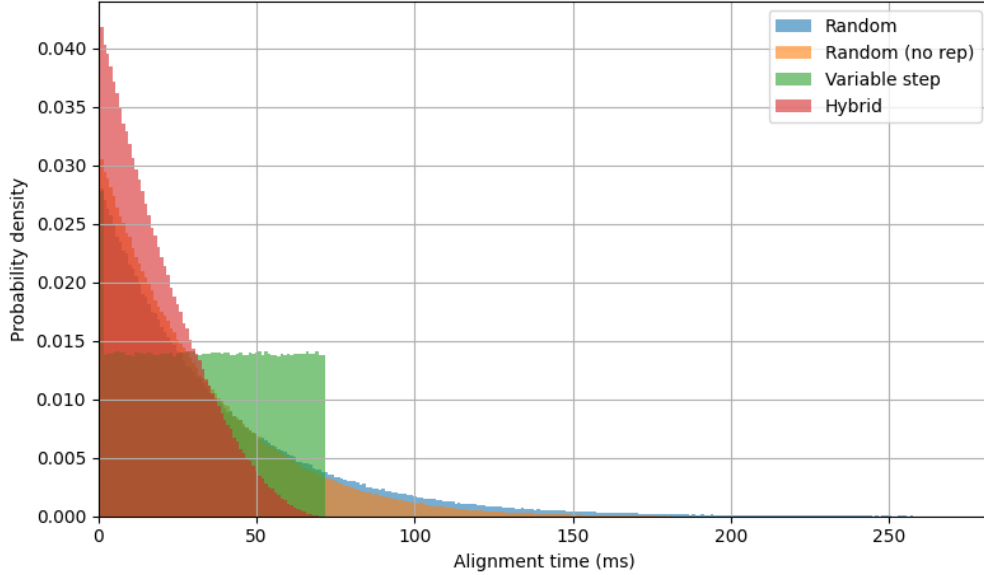


Figure 15: PDF of Beam Alignment Time in 2D for S3, S4, S5 & S6 ($\alpha = 10^\circ$)

3.4.3 3D Results: Mean Alignment Time

We next evaluate the mean alignment time in the 3D scenario, where transmitter misalignment occurs in both azimuth and elevation. The simulations were conducted for beamwidth values $\alpha \in \{5^\circ, 10^\circ, 20^\circ, 30^\circ, 60^\circ, 90^\circ\}$. Figure. 16 presents the mean alignment time for the sequential scanning strategies (S1 and S2), evaluated using $N_{\text{trials}} = 10,000$ Monte Carlo trials per beamwidth. Similar to the 2D case, clockwise and counter-clockwise scanning shows nearly identical mean performance, confirming that scan direction has no impact

on the average delay. However, the overall alignment time in 3D is significantly larger than in 2D due to the substantially increased angular search space.

Figure. 17 compares the mean alignment time of the randomised and structured strategies (S3–S6), evaluated using $N_{\text{trials}} = 100,000$ Monte Carlo trials per beamwidth. A key observation is that random probing with repetition (S3) and random probing without repetition (S4) achieve nearly identical performance across all evaluated beamwidth values. This indicates that in 3D, repetition avoidance provides limited benefit since the search space is large and the probability of selecting the same direction is relatively low. In contrast, structured grid-based strategies (S5 and S6) significantly reduce mean alignment time, particularly for narrow beams.

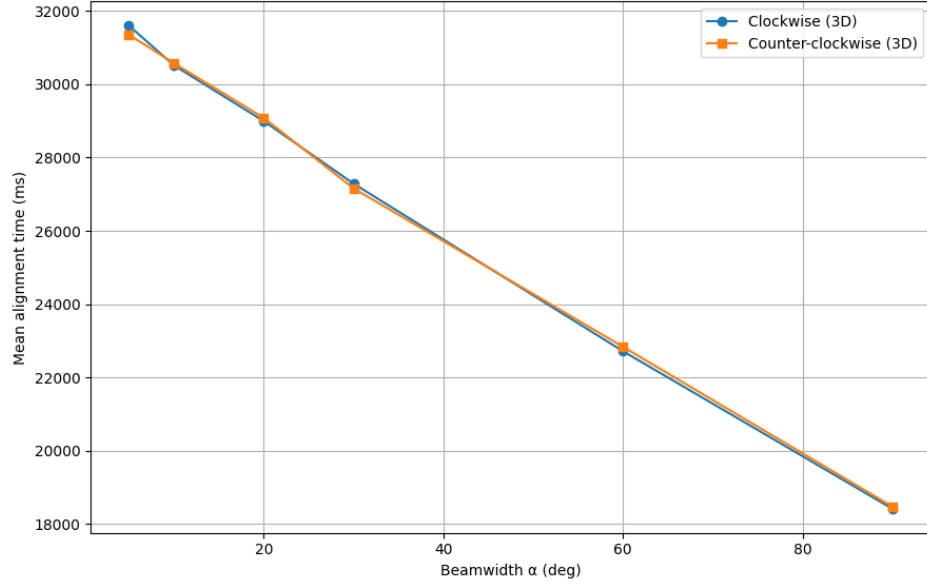


Figure 16: Mean Alignment Time vs Beamwidth (3D) for S1 & S2

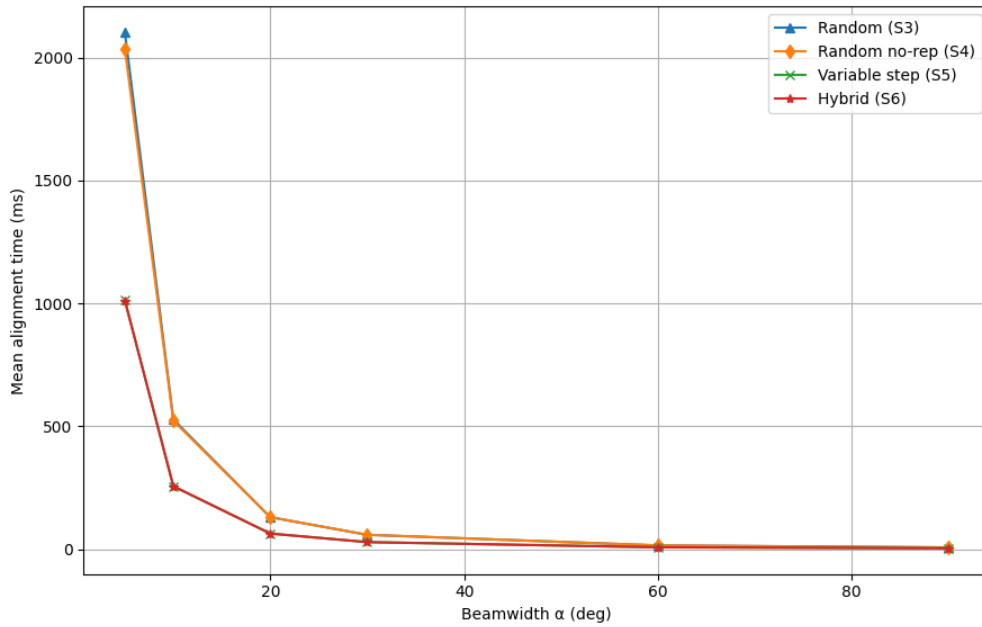


Figure 17: Mean Alignment Time vs Beamwidth (3D) for S3, S4, S5 & S6

3.4.4 3D Results: Probability Density Function

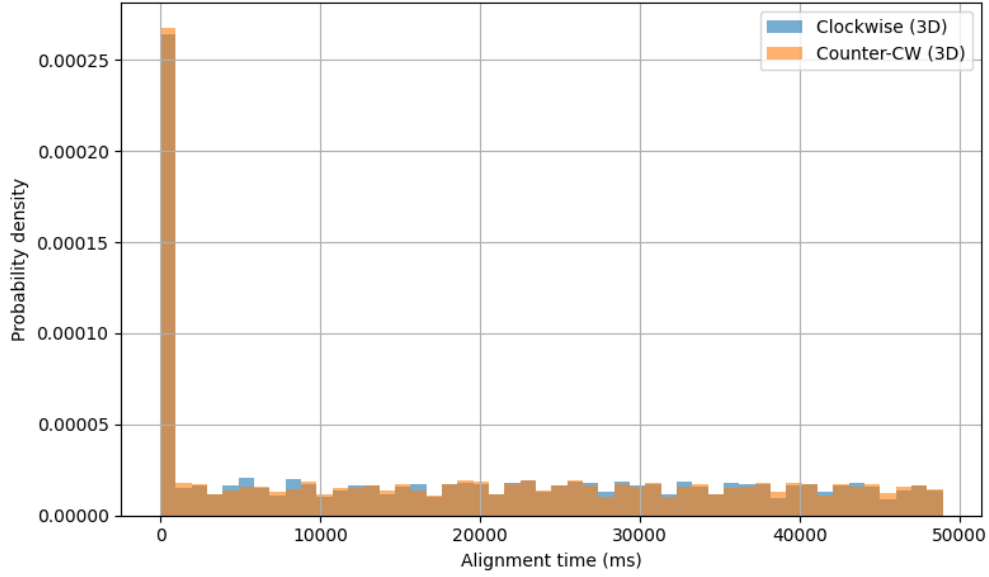


Figure 18: PDF of Beam Alignment Time in 3D for S1 & S2 ($\alpha = 90^\circ$)

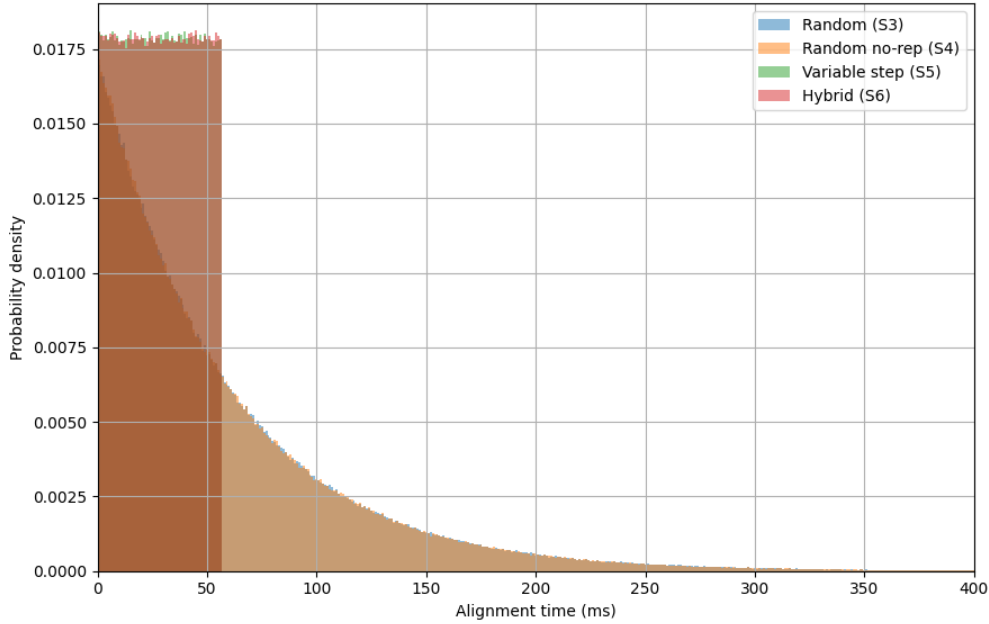


Figure 19: PDF of Beam Alignment Time in 3D for S3, S4, S5 & S6 ($\alpha = 30^\circ$)

In the 3D scenario, we evaluate the probability density function (PDF) of the alignment time. Figure. 18 shows the PDF for the sequential scanning strategies (S1 and S2). Similar to 2D, the distributions are spread almost uniformly over a wide range of alignment times, which results in highly unpredictable delays and large worst-case alignment times.

Figure. 19 presents the PDF comparison for strategies S3–S6 for $\alpha = 30^\circ$. The random probing strategies (S3 and S4) exhibit long-tailed PDFs, indicating that although alignment can occur quickly in many trials, there remains a non-negligible probability of very large alignment delays. Moreover, the PDFs of S3 and

S4 are nearly identical, confirming that avoiding repetition provides limited improvement in 3D due to the much larger search space.

In contrast, the structured strategies (S5 and S6) generated bounded PDFs that are concentrated at low alignment times, indicating faster and more reliable alignment with significantly reduced worst-case delay. Overall, the PDF results highlight that in 3D, reducing the effective search space using a beam grid dominates performance improvements.

3.5 Assumptions and Limitations

This analysis is based on a simplified beam alignment model. The receiver is assumed to remain perfectly aligned toward the transmitter. Antennas are modelled using an ideal cone-shaped radiation pattern with constant gain inside the main lobe and zero gain outside, neglecting side lobes and back lobes. Alignment detection is assumed to be instantaneous once beam overlap occurs, with each probing attempt costing exactly 1 ms. These assumptions enable comparison of beam-search strategies, but should be considered when interpreting the results in practical sub-THz deployments.

3.6 Conclusion

Results show that increasing the beamwidth α significantly reduces alignment time for all strategies, since wider beams increase the probability of Tx–Rx overlap. However, in practical sub-THz links, large beamwidths are not always feasible because they reduce directional antenna gain and thus degrade the link budget and can limit achievable range and throughput. Furthermore, the 3D results highlight that geometry-aware strategies based on beam grids are essential, as random probing provides limited benefit in the much larger spherical search space. Overall, efficient beam search strategies are critical for sub-THz systems, especially in 3D alignment.

4 Part 4: Near-Field Setup

4.1 Motivation

Sub-terahertz (sub-THz) frequency bands are important for future wireless systems because they offer very large bandwidths. At the same time, the free-space path loss is high at these frequencies, which motivates the use of large antenna arrays to obtain sufficient beamforming gain. As the array aperture becomes large and the wavelength becomes small, many practical communication distances fall into the *radiative near-field* (Fresnel) region rather than the far-field region.

In the radiative near-field, the plane-wave assumption is no longer accurate because the wavefront is better described as spherical. This causes the phase to vary across the array aperture, which cannot be captured well by standard far-field channel models. These near-field effects are therefore expected to be important for large planar arrays operating at sub-THz frequencies.

A commonly used reference boundary between near-field and far-field operation is the Fraunhofer distance. However, analytical expressions for the Fraunhofer distance are typically obtained from simplified geometric assumptions and may be conservative. Recent works motivate evaluating the near-field boundary using a physically meaningful phase criterion, based on the maximum phase deviation across the array aperture. Motivated by this, this work studies the near-field boundary for *unequal* planar antenna arrays using both an analytical Fraunhofer distance expression and an empirical, phase-based evaluation.

4.2 System Model and Assumptions

We consider a line-of-sight (LoS) communication scenario consisting of a square planar *transmit* (Tx) antenna array and a square planar *receive* (Rx) antenna array operating at a carrier frequency f in the sub-THz band. The Tx array comprises $N_1 \times N_1$ antenna elements, while the Rx array comprises $N_2 \times N_2$ antenna elements, with $N_2 \leq N_1$. The two arrays are parallel to each other and separated by a center-to-center distance d .

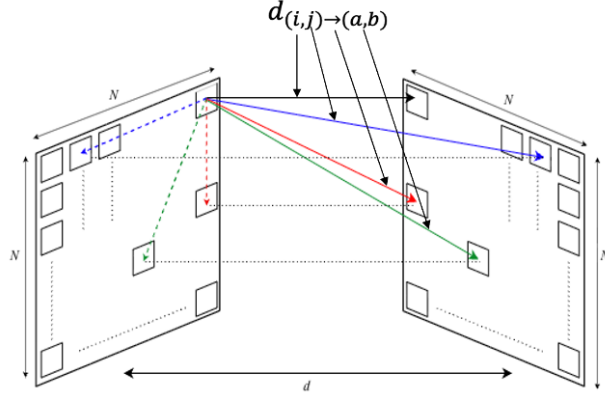


Figure 20: Empirical boundary distance d_T and analytical Fraunhofer distance d_F as functions of array size for different values of N_1 and N_2 .

Each antenna element is modeled as an isotropic point radiator transmitting a single-tone sinusoidal signal. The inter-element spacing for both arrays is chosen as

$$\Delta = \frac{\lambda}{2}, \quad (15)$$

where $\lambda = \frac{c}{f}$ denotes the wavelength and c is the speed of light. This choice of spacing is widely used for array processing and avoids spatial aliasing.

Since the arrays can be electrically large, using a single free-space path loss (FSPL) model between the Tx and Rx array centers is not accurate in the radiative near-field region. However, individual antenna elements are electrically small, and therefore propagation can be modeled on a per-element basis. Consequently, each Tx–Rx antenna element pair is modeled as an individual propagation path with its own distance-dependent phase.

A Cartesian coordinate system is used where the arrays lie in parallel planes and their centers are aligned. The positions of the antenna elements are determined by their element indices and the spacing Δ . This setup enables direct computation of the path length differences across the arrays, which is required to quantify near-field phase variations.

4.3 Phase-Based Near-Field Characterization

In the radiative near-field region, spherical wave propagation results in non-uniform phase variations across the receiving array. To quantify when the far-field plane-wave approximation becomes valid, the maximum phase deviation across the array aperture is evaluated.

The shortest propagation path is the center-to-center distance between the Tx and Rx arrays:

$$R_{\min} = d. \quad (16)$$

The longest propagation path occurs between diagonally opposite corner elements of the Tx and Rx arrays. With half-wavelength spacing, the maximum element displacement from the array center (along one axis in the array plane) is

$$x_1 = \frac{N_1 - 1}{2} \Delta, \quad x_2 = \frac{N_2 - 1}{2} \Delta. \quad (17)$$

The corresponding longest path length is then

$$R_{\max}(d) = \sqrt{d^2 + 2(x_1 + x_2)^2}. \quad (18)$$

Using these path lengths, the maximum unwrapped phase difference across the aperture is expressed as

$$\Delta\phi_{\max}(d) = \frac{2\pi}{\lambda} (R_{\max}(d) - R_{\min}(d)). \quad (19)$$

This quantity provides a direct measure of the deviation from a plane-wave model. In the far-field region, $\Delta\phi_{\max}(d)$ becomes small, while in the near-field region it can be large, indicating significant spherical wave effects.

For visualization, the wrapped phase spread $\Delta\phi_{\max}(d) \bmod 2\pi$ is also plotted, as it highlights the periodic nature of phase while preserving the overall distance-dependent trend.

4.4 Empirical Fraunhofer Distance Based on Phase Criterion

A practical boundary between radiative near-field and far-field operation can be defined by limiting the maximum phase deviation across the array aperture. In this work, the far-field condition is assumed to be satisfied when the maximum phase spread does not exceed a prescribed threshold:

$$\Delta\phi_{\max}(d) \leq \frac{\pi}{8}. \quad (20)$$

Based on this criterion, the *empirical near-field boundary distance*, denoted by d_T , is defined as the smallest distance for which the phase constraint is satisfied:

$$d_T = \min \left\{ d : \Delta\phi_{\max}(d) \leq \frac{\pi}{8} \right\}. \quad (21)$$

The value of d_T is obtained numerically by evaluating $\Delta\phi_{\max}(d)$ over a logarithmically spaced distance grid and selecting the distance at which the phase threshold is first met. This empirical approach directly enforces the physical phase condition and accounts for the exact array geometry and spherical wave propagation.

4.5 Analytical Fraunhofer Distance for Unequal Planar Arrays

For comparison with the empirical, phase-based boundary, an analytical expression for the Fraunhofer distance is also considered. The Fraunhofer distance provides a conventional estimate of the minimum separation distance beyond which the plane-wave approximation is expected to hold across the antenna apertures.

For two square planar arrays of unequal sizes, the Fraunhofer distance can be written as

$$d_F = \frac{4}{\lambda} (D_1^2 + D_2^2), \quad (22)$$

where D_1 and D_2 denote the physical aperture sizes of the transmit and receive arrays, respectively.

With half-wavelength element spacing, the physical side length of an $N_i \times N_i$ planar array is

$$D_i = (N_i - 1) \frac{\lambda}{2} \approx \frac{N_i \lambda}{2}, \quad (23)$$

where the approximation holds for sufficiently large N_i .

This analytical expression is derived under simplifying geometric assumptions and provides a conservative estimate of the far-field boundary. It does not explicitly account for the exact element-wise path geometry, but it serves as a useful reference for evaluating the extent of the radiative near-field region.

4.6 Numerical Results

This subsection presents numerical results illustrating the behavior of the radiative near-field boundary for unequal planar antenna arrays. All results are obtained using the phase-based formulation described in the previous subsections.

4.6.1 Phase Difference Versus Distance

Fig. 21 shows the wrapped maximum phase difference $\Delta\phi_{\max}(d) \bmod 2\pi$ as a function of the separation distance d . At short distances, the phase difference exhibits multiple 2π wraps, indicating strong near-field behavior. As the distance increases, the phase difference decreases and eventually falls below the $\pi/8$ threshold.

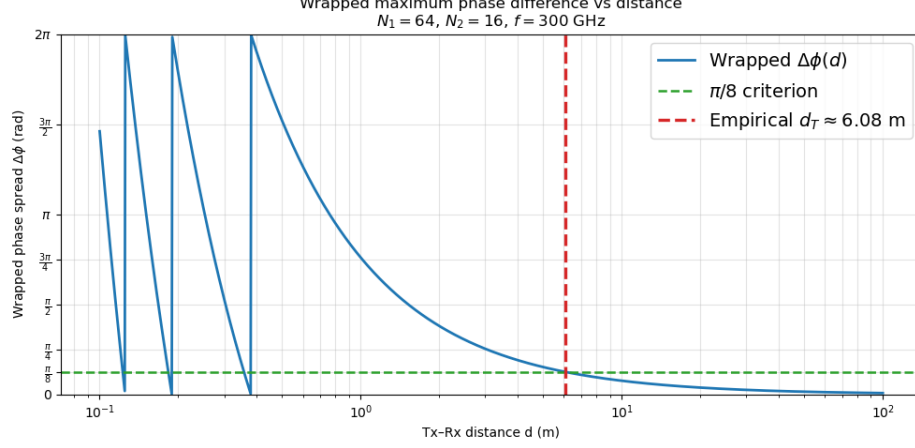


Figure 21: Wrapped maximum phase difference $\Delta\phi_{\max}(d) \bmod 2\pi$ versus separation distance d for a fixed array configuration.

The distance at which this threshold is crossed corresponds to the empirical boundary d_T . For reference, the analytical Fraunhofer distance d_F is also indicated in the figure.

4.6.2 Dependence on Array Size

To study the impact of array size, the near-field boundary is evaluated for different values of N_1 and N_2 . Fig. 23 shows that both the empirical boundary d_T and the analytical Fraunhofer distance d_F increase rapidly as the number of antenna elements increases. This behavior is expected, since larger arrays have larger physical apertures, which lead to increased phase variations across the array for a given distance.

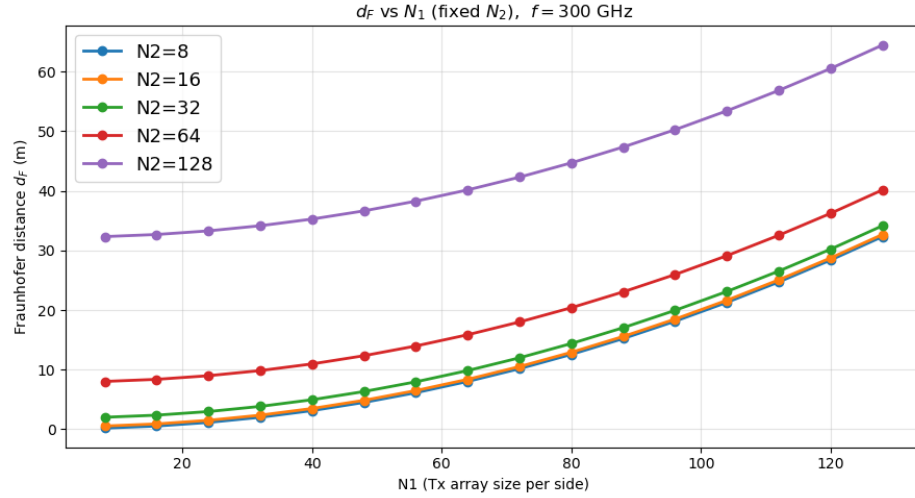


Figure 22: Empirical boundary distance d_T and analytical Fraunhofer distance d_F as functions of array size for different values of N_1 and N_2 .

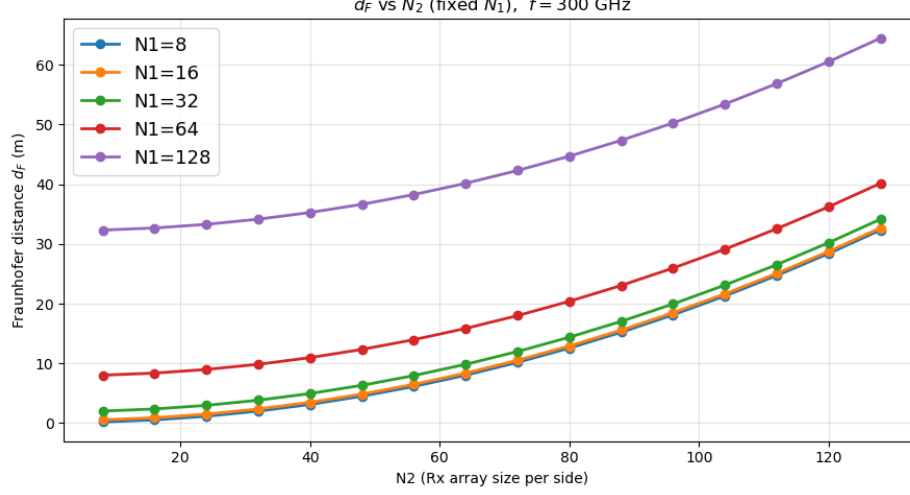


Figure 23: Empirical boundary distance d_T and analytical Fraunhofer distance d_F as functions of array size for different values of N_1 and N_2 .

When one array size is fixed, increasing the size of the other array still leads to a noticeable increase in the near-field boundary, highlighting the importance of considering unequal array configurations.

4.6.3 Dependence on Carrier Frequency

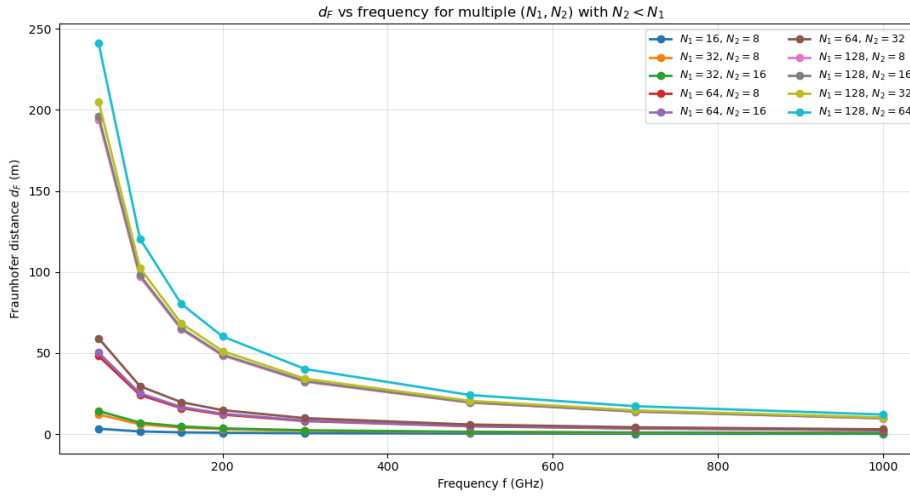


Figure 24: Empirical boundary distance d_T and analytical Fraunhofer distance d_F versus carrier frequency for fixed array sizes.

The dependence of the near-field boundary on the carrier frequency is also investigated. For fixed array sizes, both d_T and d_F decrease as the carrier frequency increases. This trend follows from the reduction in wavelength at higher frequencies, which reduces the physical aperture measured in wavelengths.

Despite this decrease, the results show that the radiative near-field region can still extend over several meters at sub-THz frequencies when large antenna arrays are employed.

4.7 Comparison Between Empirical and Analytical Boundaries

A comparison between the empirical boundary d_T and the analytical Fraunhofer distance d_F shows that the two values are generally of the same order of magnitude, but not identical. In all evaluated scenarios, the empirical boundary is smaller than the analytical Fraunhofer distance.

This difference arises because the analytical expression is conservative and based on simplified geometric assumptions, whereas the empirical boundary directly enforces a phase-error criterion using the exact element-wise path geometry. As a result, d_T represents the minimum distance required to satisfy the phase condition, while d_F provides a safe upper bound for far-field operation.

4.8 Key Observations

From the numerical analysis, the following observations can be made:

- Large planar antenna arrays operating at sub-THz frequencies exhibit an extensive radiative near-field region.
- The near-field boundary increases rapidly with array size and decreases with increasing carrier frequency.
- Unequal array configurations still lead to significant near-field effects and should not be neglected.
- The empirical, phase-based boundary provides a physically meaningful and geometry-consistent characterization of the near-field region.
- Analytical Fraunhofer distance expressions serve as conservative reference values and align well with empirical trends when consistent aperture definitions are used.

4.9 Summary

In this part, the radiative near-field behavior of unequal planar antenna arrays was analyzed using both analytical and empirical approaches. By explicitly modeling element-wise propagation paths and evaluating the maximum phase deviation across the array aperture, an empirical near-field boundary was obtained and compared with a conventional Fraunhofer distance expression.

The results confirm that near-field effects are unavoidable for large antenna arrays at sub-THz frequencies and must be considered in system modeling and design. The presented analysis provides insight into how array size asymmetry and carrier frequency influence the extent of the radiative near-field region.

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